27-7-26

Experiment -1

Basic Networking Commands in linux and Windows.

AIM:

To study and execute basic networking commands available in linux and windows OS for diagnosing and troubleshooting network in use.

REQUIREMENTS:

PC with windows/ linux, Terminal or command Prompt Active network connection.

PROCEDURE:

1) Open command prompt (windows) or terminal
2) Execute 'ipconfig' or 'ifconfig' to view ip configuration
3) Use ping 8.8.8.8 to test connectivity to google DNS server.
4) Execute 'tracer google.com' or 'tracer route google.com' to view path packet
5) Run netstat -an to list active network connections and listening ports
6) Use 'nslookup google.com' to perform DNS lookup.

commands

1) ipconfig /all # windows full ip configuration
2) ip a # linux ip address info
3) ping -c 4 8.8.8.8 # linux ping 4 packets
4) ping 8.8.8.8 # windows ping
5) traceroute google.com # linux - trace route
6) netstat - tuln # linux - listen port
7) nslookup ~~configuration~~ google.com # DNS resolution
8) arp - a # arp table
9) route -n # Routing table (linux)

7) Execute 'arp -a' to view the ARP cache
8) use 'route print' (windows) or 'route -n' (linux) to view the routing table
9) Record all output and analyze the results.

RESULT:

31/8/25

Basic networking commands were successfully executed on both linux and windows. IP addresses, routing tables, ARP caches and DNS resolution were observed and recorded.